eedi-mining-misconceptions-in-mathematics

12位 Solution

Rank	12
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Version	Japanese Translation

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551429
Retrieved with Kaggle CLI on 2026-10-21.

Kaggle Discussionの主投稿と、技術的補足を含む選定済みQ&Aを日本語へ翻訳した版です。code、URL、model名、識別子、数値は原文を保持しています。

このコンペを開催してくださったKaggleプラットフォームとEdiに感謝を申し上げます。このコンペを通じて、recallとrankingの技術について、特に大規模modelをembeddingのcontrastive learningとrerankingに活用する方法について多くを学びました。最終的にsoloで金メダルを獲得し、Grand Masterに一歩近づきました。

概要

主な戦略は、Retrieverとrerankerの組み合わせです。

合成データ生成

Chain of Thought（CoT）のアプローチを用いて、misconceptionに基づき4回に分けて合成データを生成し、合計で10,000件を超える外部データを得ました。promptは次のとおりです:

```
You are an intelligent math problem-solving expert.
I will provide you with a misconception-related error reason. Based on this error, generate a math single-choice question along with four options, where only one is correct, and the other three are distractors.
The error reason provided corresponds to one of the incorrect options among the three distractors.
Below are several examples:
Example 1:
Input: Confuses the order of operations, believes addition comes before multiplication
Output: [question]ConstructName:Use the order of operations to carry out calculations involving powersSubjectName:BIDMASQuestionDesc:[3 times 2+4-5]Where do the brackets need to go to make the answer equal ( 13 ) ?[Candidate answers]A.( 3 times(2+4)-5 )B.( 3 times 2+(4-5) )C.( 3 times(2+4-5) )D.Does not need brackets[Correct Answer]( 3 times(2+4)-5 )[Misconcepte Incorrect answer]Does not need brackets

Example 2:
Input: Does not know that to factorise a quadratic expression, to find two numbers that add to give the coefficient of the x term, and multiply to give the non-variable term
Output: [question]ConstructName:Simplify an algebraic fraction by factorising the numeratorSubjectName:Simplifying Algebraic FractionsQuestionDesc:Simplify the following, if possible: ( frac{m^{2}+2 m-3}{m-3} )[Candidate answers]A.( m+1 )B.( m+2 )C.( m-1 )D.Does not simplify[Correct Answer]Does not simplify[Misconcepte Incorrect answer]( m+1 )

Example 3:
Input: Thinks that when you cancel identical terms from the numerator and denominator, they just disappear
Output: [question]ConstructName:Simplify an algebraic fraction by factorising the numeratorSubjectName:Simplifying Algebraic FractionsQuestionDesc:Simplify the following, if possible: ( frac{m^{2}+2 m-3}{m-3} )[Candidate answers]A.( m+1 )B.( m+2 )C.( m-1 )D.Does not simplify[Correct Answer]Does not simplify[Misconcepte Incorrect answer]( m+2 )

Example 4:
Input: Does not know that to factorise a quadratic expression, to find two numbers that add to give the coefficient of the x term, and multiply to give the non-variable term
Output: [question]ConstructName:Simplify an algebraic fraction by factorising the numeratorSubjectName:Simplifying Algebraic FractionsQuestionDesc:Simplify the following, if possible: ( frac{m^{2}+2 m-3}{m-3} )[Candidate answers]A.( m+1 )B.( m+2 )C.( m-1 )D.Does not simplify[Correct Answer]Does not simplify[Misconcepte Incorrect answer]( m-1 )
Please generate similar questions and options based on the cases above.
You first need to think about the reasoning logic according to the chain of thought, then provide the specific case.
The output format is:
```

```
<thought>Insert your thoughts on the error reason here</thought>
<output>Insert your question and options here</output>
```

外部データにより、recallが0.430から0.470に改善しました（single fold）。

Retriever

全体的なアプローチとしては、まずSFR modelで100個の候補をretrieveし、その後Qwen modelでcontrastive learningを学習しました。3つのmodelのensembleは、embeddingを連結することで実現しました: `emb=concat(emb1,emb2,emb3)`。

- Qwen2.5-32b-instruct-awqで潜在的なerrorを推論し、元のqueryと組み合わせてQwen2.5-14B-instructを学習しました。5-foldのLoRA weightを平均して`emb1`を得ました。
- 元のqueryでQwen2.5-14B-instructを学習し、5-foldのLoRA weightを平均して`emb2`を得ました。
- notebookのopen-source modelを`emb3`に使いました。
- Public score: 0.526、Private score: 0.485

Reranker

Retrieverがretrieveしたtop 50の候補（5 foldのoofから得た結果）について、毎回25個をランダムに選び、25-class予測を行いました。最終スコアは、3つのmodelの確率を重み付けしてensembleすることで達成しました:

- Public score: 0.597、Private score: 0.564
- Qwen2.5-32b-instruct-awq origin query + vllm infer: 5-foldのLoRA weightを平均、Public: 0.570
- Qwen2.5-14b-instruct origin query: 5-foldのLoRA weightを平均、Public: 0.567
- Qwen2.5-14b-instruct query + reason: 5-foldのLoRA weightを平均、Public: 0.558

うまくいかなかったこと

- Qwen2.5-coder
- oofに対する後段rerankerとしてのLGB/CatBoost
- Reranker query + misconception last_pool => 0/1